

# Lecture 3: Basic Matrix Operations

MATH 2308

January 18-20, 2023

# Matrices are interesting in their own rights

So far, we've only used matrices as shorthand for an underlying linear system. It turns out that matrices have useful algebraic properties, which we will now study.

# The dimension of a matrix

The **dimension** or size of a matrix is the number of rows and columns. An  $m \times n$  matrix has  $m$  rows (horizontal) and  $n$  columns (vertical). Here are some examples.

$$\begin{bmatrix} 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 3 & 6 & 2 \\ 2 & 1 & 0 & 0 \\ 3 & 5 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$

# The dimension of a matrix

The **dimension** or size of a matrix is the number of rows and columns. An  $m \times n$  matrix has  $m$  rows (horizontal) and  $n$  columns (vertical). Here are some examples.

$$\begin{bmatrix} 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix} \text{ is } 2 \times 3$$

$$\begin{bmatrix} -1 & 3 & 6 & 2 \\ 2 & 1 & 0 & 0 \\ 3 & 5 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$

# The dimension of a matrix

The **dimension** or size of a matrix is the number of rows and columns. An  $m \times n$  matrix has  $m$  rows (horizontal) and  $n$  columns (vertical). Here are some examples.

$$\begin{bmatrix} 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix} \text{ is } 2 \times 3$$

$$\begin{bmatrix} -1 & 3 & 6 & 2 \\ 2 & 1 & 0 & 0 \\ 3 & 5 & 1 & 2 \end{bmatrix} \text{ is } 3 \times 4$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$

# The dimension of a matrix

The **dimension** or size of a matrix is the number of rows and columns. An  $m \times n$  matrix has  $m$  rows (horizontal) and  $n$  columns (vertical). Here are some examples.

$$\begin{bmatrix} 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix} \text{ is } 2 \times 3$$

$$\begin{bmatrix} -1 & 3 & 6 & 2 \\ 2 & 1 & 0 & 0 \\ 3 & 5 & 1 & 2 \end{bmatrix} \text{ is } 3 \times 4$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \text{ is } 2 \times 2 \text{ (a **square matrix**)}$$

# Row and column vectors

An  $m \times 1$  matrix is called a **column vector**.

$$\begin{bmatrix} 2 \\ 4 \\ -1 \\ 6 \\ 7 \end{bmatrix}$$

A  $1 \times n$  matrix is called a **row vector**.

$$[1 \quad 2 \quad -1 \quad 5 \quad 4 \quad 8]$$

# Matrix addition

Two matrices of the same dimension can be added together. The addition is entry-wise.

$$\begin{bmatrix} 2 & 4 & -1 \\ 3 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 4 & 2 & 1 \\ 2 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 2+4 & 4+2 & -1+1 \\ 3+2 & 2+(-2) & 3+3 \end{bmatrix} \\ = \begin{bmatrix} 6 & 6 & 0 \\ 5 & 0 & 6 \end{bmatrix}$$



# Matrix subtraction

Matrix subtraction is similar. Again, the matrices must be of the same size.

$$\begin{bmatrix} 2 & 4 & -1 \\ 3 & 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 2 & 1 \\ 2 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 2-4 & 4-2 & -1-1 \\ 3-2 & 2-(-2) & 3-3 \end{bmatrix} \\ = \begin{bmatrix} -2 & 2 & -2 \\ 1 & 4 & 0 \end{bmatrix}$$

# Scalar Multiplication

A matrix can be multiplied by a constant. This process is called **scalar multiplication** to avoid confusion with **matrix multiplication**, which we will see in a bit. In scalar multiplication, each entry in the matrix is multiplied by the constant.

# Scalar Multiplication

A matrix can be multiplied by a constant. This process is called **scalar multiplication** to avoid confusion with **matrix multiplication**, which we will see in a bit. In scalar multiplication, each entry in the matrix is multiplied by the constant.

For instance

$$2 \begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2 & 2 \cdot 1 \\ 2 \cdot (-3) & 2 \cdot 4 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ -6 & 8 \end{bmatrix}.$$

# Scalar Multiplication

A matrix can be multiplied by a constant. This process is called **scalar multiplication** to avoid confusion with **matrix multiplication**, which we will see in a bit. In scalar multiplication, each entry in the matrix is multiplied by the constant.

For instance

$$2 \begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2 & 2 \cdot 1 \\ 2 \cdot (-3) & 2 \cdot 4 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ -6 & 8 \end{bmatrix}.$$

The **additive inverse** of a matrix  $\mathbf{A}$  is  $-\mathbf{A} = (-1)\mathbf{A}$ .

# The zero matrix and properties of matrix arithmetic

The  $m \times n$  zero matrix  $\mathbf{0}$  is the  $m \times n$  matrix with all zero entries. For instance, the  $2 \times 3$  zero matrix is

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

# The zero matrix and properties of matrix arithmetic

The  $m \times n$  zero matrix  $\mathbf{0}$  is the  $m \times n$  matrix with all zero entries. For instance, the  $2 \times 3$  zero matrix is

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

## Properties of matrix addition

Let  $\mathbf{A}$ ,  $\mathbf{B}$ , and  $\mathbf{C}$  be any  $m \times n$  matrices, and let  $\mathbf{0}$  be the  $m \times n$  zero matrix. Then

- Commutative property:  $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
- Associative property:  $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$
- Identity property of addition:  $\mathbf{A} + \mathbf{0} = \mathbf{0} + \mathbf{A} = \mathbf{A}$
- Inverse property of addition:  $\mathbf{A} + (-\mathbf{A}) = (-\mathbf{A}) + \mathbf{A} = \mathbf{0}$

# Matrix multiplication: A motivating example

A baseball team had 7000 fans in attendance, and total ticket receipts were \$26,400. Box seats cost \$6, grandstand seats cost \$4, and bleacher seats cost \$2. If the number of box seats sold was one-third the number of bleacher seats, how many of each seat were sold?

# Matrix multiplication: A motivating example

A baseball team had 7000 fans in attendance, and total ticket receipts were \$26,400. Box seats cost \$6, grandstand seats cost \$4, and bleacher seats cost \$2. If the number of box seats sold was one-third the number of bleacher seats, how many of each seat were sold?

Let  $x$  be the number of box seats,  $y$  be the number of grandstand seats, and let  $z$  be the number of bleacher seats. Our equations are then

① Total tickets:  $x + y + z = 7000$



# Matrix multiplication: A motivating example

A baseball team had 7000 fans in attendance, and total ticket receipts were \$26,400. Box seats cost \$6, grandstand seats cost \$4, and bleacher seats cost \$2. If the number of box seats sold was one-third the number of bleacher seats, how many of each seat were sold?

Let  $x$  be the number of box seats,  $y$  be the number of grandstand seats, and let  $z$  be the number of bleacher seats. Our equations are then

- 1 Total tickets:  $x + y + z = 7000$
- 2 Revenue:  $6x + 4y + 2z = 26400$

# Matrix multiplication: A motivating example

A baseball team had 7000 fans in attendance, and total ticket receipts were \$26,400. Box seats cost \$6, grandstand seats cost \$4, and bleacher seats cost \$2. If the number of box seats sold was one-third the number of bleacher seats, how many of each seat were sold?

Let  $x$  be the number of box seats,  $y$  be the number of grandstand seats, and let  $z$  be the number of bleacher seats. Our equations are then

- 1 Total tickets:  $x + y + z = 7000$
- 2 Revenue:  $6x + 4y + 2z = 26400$
- 3 Box/Bleacher:  $3x = z$  or  $3x - z = 0$ .

# The solution

Here is the augmented matrix and its reduced row echelon form.

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 7000 \\ 6 & 4 & 2 & 26400 \\ 3 & 0 & -1 & 0 \end{array} \right], \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 400 \\ 0 & 1 & 0 & 5400 \\ 0 & 0 & 1 & 1200 \end{array} \right]$$

# The solution

Here is the augmented matrix and its reduced row echelon form.

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 7000 \\ 6 & 4 & 2 & 26400 \\ 3 & 0 & -1 & 0 \end{array} \right], \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 400 \\ 0 & 1 & 0 & 5400 \\ 0 & 0 & 1 & 1200 \end{array} \right]$$

There were 400 box seats sold, 5400 grandstand seats, and 1200 bleacher seats.

# Motivation, continued

Our solution again is 400 box seats, 5400 grandstand seats, and 1200 bleacher seats. We can store these numbers in a column vector

$$\begin{bmatrix} 400 \\ 5400 \\ 1200 \end{bmatrix}.$$

# Matrix multiplication

The coefficient matrix and solution vector are again,

$$\begin{bmatrix} 1 & 1 & 1 \\ 6 & 4 & 2 \\ 3 & 0 & -1 \end{bmatrix} \text{ and } \begin{bmatrix} 400 \\ 5400 \\ 1200 \end{bmatrix}.$$

The total revenue can be calculated from the solution vector and the second row:

$$\text{Revenue} = 6 \times 400 + 4 \times 5400 + 2 \times 1200 = \$26,400.$$

# Matrix multiplication

The coefficient matrix and solution vector are again,

$$\begin{bmatrix} 1 & 1 & 1 \\ 6 & 4 & 2 \\ 3 & 0 & -1 \end{bmatrix} \text{ and } \begin{bmatrix} 400 \\ 5400 \\ 1200 \end{bmatrix}.$$

The total revenue can be calculated from the solution vector and the second row:

$$\text{Revenue} = 6 \times 400 + 4 \times 5400 + 2 \times 1200 = \$26,400.$$

You might recognize this as a **dot product** of vectors:

$$(a_1, a_2, a_3) \cdot (b_1, b_2, b_3) = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

# Matrix multiplication, continued

When we do the same thing with the first row, we get

$$1 \times 400 + 1 \times 5400 + 1 \times 1200 = 7000,$$

which tells us the number of tickets sold.



# Matrix multiplication, continued

When we do the same thing with the first row, we get

$$1 \times 400 + 1 \times 5400 + 1 \times 1200 = 7000,$$

which tells us the number of tickets sold.

The dot product with the third row is

$$3 \times 400 + 0 \times 5400 + (-1) \times 1200 = 0,$$

which tells us the relationship between the number of box seats and bleacher seats sold.

# Multiplying a vector by a matrix

We can put these three dot products into their own vector, and we see our first matrix product.

$$\begin{bmatrix} 1 & 1 & 1 \\ 6 & 4 & 2 \\ 3 & 0 & -1 \end{bmatrix} \begin{bmatrix} 400 \\ 5400 \\ 1200 \end{bmatrix} = \begin{bmatrix} 7000 \\ 26400 \\ 0 \end{bmatrix}$$

# Multiplying a vector by a matrix

We can put these three dot products into their own vector, and we see our first matrix product.

$$\begin{bmatrix} 1 & 1 & 1 \\ 6 & 4 & 2 \\ 3 & 0 & -1 \end{bmatrix} \begin{bmatrix} 400 \\ 5400 \\ 1200 \end{bmatrix} = \begin{bmatrix} 7000 \\ 26400 \\ 0 \end{bmatrix}$$

The first entry in the product vector is the dot product of the first row of the matrix with the column vector. The second entry is the dot product of the second row with the vector, and so on. This is how we define the multiplication of a column vector by a matrix.

# Multiplying a vector by a matrix, continued

We can multiply a (column) vector on the left by a matrix whenever the length of the vector matches the number of columns in the matrix. The length of the product will be the number of rows in the matrix.

$$\underbrace{\begin{bmatrix} 2 & 4 & 1 & 5 \\ 1 & -3 & 2 & 4 \end{bmatrix}}_{2 \times 4} \underbrace{\begin{bmatrix} 3 \\ -1 \\ 2 \\ 3 \end{bmatrix}}_{4 \times 1} = \begin{bmatrix} 2 \cdot 3 + 4 \cdot (-1) + 1 \cdot 2 + 5 \cdot 3 \\ 1 \cdot 3 + (-3) \cdot (-1) + 2 \cdot 2 + 4 \cdot 3 \end{bmatrix}$$
$$= \underbrace{\begin{bmatrix} 19 \\ 22 \end{bmatrix}}_{2 \times 1}$$

# Multiplying two matrices

Having defined what it means to multiply a vector by a matrix, we can now talk about multiplying two matrices.

$$\underbrace{\begin{bmatrix} 2 & 4 & 1 & 5 \\ 1 & -3 & 2 & 4 \end{bmatrix}}_{2 \times 4} \underbrace{\begin{bmatrix} 1 & 2 \\ -2 & 3 \\ 2 & 2 \\ -1 & 4 \end{bmatrix}}_{4 \times 2} = \underbrace{\begin{bmatrix} -9 & 38 \\ 7 & 13 \end{bmatrix}}_{2 \times 2}$$

# Multiplying two matrices

Having defined what it means to multiply a vector by a matrix, we can now talk about multiplying two matrices.

$$\underbrace{\begin{bmatrix} 2 & 4 & 1 & 5 \\ 1 & -3 & 2 & 4 \end{bmatrix}}_{2 \times 4} \underbrace{\begin{bmatrix} 1 & 2 \\ -2 & 3 \\ 2 & 2 \\ -1 & 4 \end{bmatrix}}_{4 \times 2} = \underbrace{\begin{bmatrix} -9 & 38 \\ 7 & 13 \end{bmatrix}}_{2 \times 2}$$

If  $\mathbf{A}$  is an  $m \times n$  matrix and  $\mathbf{B}$  is an  $n \times k$  matrix, then their product  $\mathbf{AB}$  is an  $m \times k$  matrix whose  $i$ th column is the product of  $\mathbf{A}$  and the  $i$ th column of  $\mathbf{B}$ .

# Example

Multiply

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

# Example

Multiply

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 + (-1) \cdot 3 & * \\ * & * \end{bmatrix}$$



# Example

Multiply

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -1 & * \\ 3 \cdot 1 + (-2) \cdot 3 & * \end{bmatrix}$$

# Example

Multiply

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 2 \cdot 2 + (-1) \cdot 4 \\ -3 & * \end{bmatrix}$$

# Example

Multiply

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ -3 & 3 \cdot 2 + (-2) \cdot 4 \end{bmatrix}$$

# Example

Multiply

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

$$\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ -3 & -2 \end{bmatrix}$$

# Matrix multiplication is not commutative

What happens when we multiply the same two matrices, but in the opposite order?

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 8 & -5 \\ 18 & -11 \end{bmatrix}.$$

We get a different answer. Matrix multiplication is *not* commutative.

# Matrix multiplication is not commutative

What happens when we multiply the same two matrices, but in the opposite order?

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 8 & -5 \\ 18 & -11 \end{bmatrix}.$$

We get a different answer. Matrix multiplication is *not* commutative. In fact,

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 4 \\ 1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 6 & 3 & 6 \\ 7 & 2 & -1 \end{bmatrix},$$

# Matrix multiplication is not commutative

What happens when we multiply the same two matrices, but in the opposite order?

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 8 & -5 \\ 18 & -11 \end{bmatrix}.$$

We get a different answer. Matrix multiplication is *not* commutative. In fact,

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 4 \\ 1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 6 & 3 & 6 \\ 7 & 2 & -1 \end{bmatrix},$$

but

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 4 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

isn't even defined.

# Properties of Matrix Multiplication

There are *some* properties of matrix multiplication that do hold.

## Properties of Matrix Multiplication

If **A**, **B**, and **C** are matrices such that all the indicated sums and products exist, then the following properties are satisfied.

❶ Associative:  $\mathbf{A(BC) = (AB)C}$

❷ Distributive:

- $\mathbf{A(B + C) = AB + AC}$

- $\mathbf{(A + B)C = AC + BC}$



# Identity matrices

The  $n \times n$  identity matrix  $\mathbf{I}$  has ones along its main diagonal and zeros everywhere else.

- $2 \times 2 : \mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

- $3 \times 3 : \mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

If  $\mathbf{A}$  is any matrix, and  $\mathbf{I}$  is the appropriately sized identity matrix, then  $\mathbf{AI} = \mathbf{A}$ . Similarly,  $\mathbf{IA} = \mathbf{A}$  for the appropriately sized  $\mathbf{I}$ .

# Inverse Matrices

An  $n \times n$  matrix  $\mathbf{A}$  is **invertible** if there is an  $n \times n$  matrix  $\mathbf{A}^{-1}$  such that

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

# Inverse Matrices

An  $n \times n$  matrix  $\mathbf{A}$  is **invertible** if there is an  $n \times n$  matrix  $\mathbf{A}^{-1}$  such that

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

For instance, If  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ , then

$$\mathbf{AB} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I},$$

and

$$\mathbf{BA} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I},$$

so  $\mathbf{B} = \mathbf{A}^{-1}$  (and  $\mathbf{A} = \mathbf{B}^{-1}$ ).

# Finding inverses

We can find the inverse of a matrix  $\mathbf{A}$  by solving the matrix equation  $\mathbf{AX} = \mathbf{I}$ . The way we do this is by row reduction with the matrix  $\mathbf{A}$  augmented by the columns of  $\mathbf{I}$ .

# Finding inverses

We can find the inverse of a matrix  $\mathbf{A}$  by solving the matrix equation  $\mathbf{A}\mathbf{X} = \mathbf{I}$ . The way we do this is by row reduction with the matrix  $\mathbf{A}$  augmented by the columns of  $\mathbf{I}$ . For example, to find  $\mathbf{A}^{-1}$  for  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ , we row reduce

$$\left[ \begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right].$$

# Finding inverses

We can find the inverse of a matrix  $\mathbf{A}$  by solving the matrix equation  $\mathbf{A}\mathbf{X} = \mathbf{I}$ . The way we do this is by row reduction with the matrix  $\mathbf{A}$  augmented by the columns of  $\mathbf{I}$ . For example, to find  $\mathbf{A}^{-1}$  for  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ , we row reduce

$$\left[ \begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right].$$

$$\begin{array}{c} \left[ \begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R1 \leftrightarrow R2} \left[ \begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array} \right] \\ \xrightarrow{-2R1 + R2 \rightarrow R2} \left[ \begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array} \right] \xrightarrow{-R2 \rightarrow R2} \left[ \begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array} \right] \end{array}$$

# Finding inverses, continued

Finally

$$\left[ \begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array} \right] \xrightarrow{-R2+R1 \rightarrow R1} \left[ \begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array} \right]$$

and our inverse is

$$\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

# Not all matrices have inverses

Not all square matrices have inverses. For instance, if  $A = \begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix}$ , when we try to find our inverse, we get

$$\left[ \begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ -2 & -4 & 0 & 1 \end{array} \right] \xrightarrow{2R_1+R_2 \rightarrow R_2} \left[ \begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{array} \right]$$

and we're stuck. This matrix has no inverse. Noninvertible matrices are also called **singular**.



# A formula for $2 \times 2$ inverses

Using row reduction on the augmented matrix

$$[\mathbf{A}|\mathbf{I}] = \left[ \begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array} \right]$$

leads to the following formula for  $\mathbf{A}^{-1}$

The inverse of a  $2 \times 2$  matrix

If  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  and  $ad - bc \neq 0$ , then  $\mathbf{A}$  is invertible and

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

# Examples

Let  $\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ .

# Examples

Let  $\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ . Then

$$\mathbf{A}^{-1} = \frac{1}{2 \cdot 4 - (-1) \cdot 3} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{11} & \frac{1}{11} \\ -\frac{3}{11} & \frac{2}{11} \end{bmatrix}.$$

# Examples

Let  $\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ . Then

$$\mathbf{A}^{-1} = \frac{1}{2 \cdot 4 - (-1) \cdot 3} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{11} & \frac{1}{11} \\ -\frac{3}{11} & \frac{2}{11} \end{bmatrix}.$$

If  $\mathbf{B} = \begin{bmatrix} 2 & 3 \\ 6 & 9 \end{bmatrix}$ ,

# Examples

Let  $\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ . Then

$$\mathbf{A}^{-1} = \frac{1}{2 \cdot 4 - (-1) \cdot 3} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 4 & 1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{11} & \frac{1}{11} \\ -\frac{3}{11} & \frac{2}{11} \end{bmatrix}.$$

If  $\mathbf{B} = \begin{bmatrix} 2 & 3 \\ 6 & 9 \end{bmatrix}$ ,

then

$$\mathbf{B}^{-1} = \frac{1}{2 \cdot 9 - 6 \cdot 3} \begin{bmatrix} 9 & -3 \\ -6 & 2 \end{bmatrix} = \frac{1}{0} \begin{bmatrix} 9 & -3 \\ -6 & 2 \end{bmatrix} \cdots$$

This matrix is not invertible.

# Higher dimensional matrices

There's no simple formula for the inverse of a  $3 \times 3$  or higher dimensional matrix, so you're either stuck with row reduction, or you can use technology.

# Properties of Inverses

The following are some helpful properties of matrix inverses.

## Properties

Let  $\mathbf{A}$  and  $\mathbf{B}$  be invertible  $n \times n$  matrices. Then

- 1  $\mathbf{A}^{-1}$  is invertible and  $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$ .
- 2  $\mathbf{AB}$  is invertible and  $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ .